



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2022

Paper: Computational Physics - I

Course Code: PHYS-3502

Roll No.

Time: 3 Hrs.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- I✓ Write a C++ program to calculate momentum and kinetic energy of the particle using functions such that $P = mv$ and $K.E. = \frac{1}{2} mv^2$.
- II✓ Write a C++ program to print diagonal elements of $C = A \times B$, where A and B are 3×3 matrices.
- III✓ Write a C++ program to print the inverse of a product of two matrices of order 3×3 .
- IV✓ Write a C++ program to calculate time period and print l(length) against T for length $L=0.2-2$, $h=0.2$ and $T = 2\pi\sqrt{l/g}$.
- V✓ Write a C++ program to calculate Area and Circumference of circle using array size=7.
- VI✓ Write a C++ program to print a 4×3 matrix by user entered values.

Q.2. Answer the following questions.

(3x10=30)

- I✓ Write a C++ program to calculate and print time against position values for Damped Harmonic Oscillator by using Euler's Method for: $a = -kx - bv/m$, given that $g = 9.8 m/s^2$, Initial position $x=0$, Initial velocity $v_i=50m/s$, Initial time $t=0$ s, $h=0.15$ s, $t_{max}=5.5s$, mass $m=1kg$, $k=$, $b=0.5$.
- II✓ Write a C++ program to find roots of equation $2x - \cos x - 3$ at $x_0 = \frac{\pi}{2}$ using Simple Iterative method (one of numerical methods)
- III✓ Write a C++ program to evaluate the integral $I = \int_0^1 e^x dx$ using Simpson's 1/3 rule for the range $[0,1]$ and $h=1/6$.

Computational Physics-I

Fifth Sem - fall 2022

• Long Questions:

(III)

Iterative Method:

→ Solution:-

$$2x - \cos x - 3 = 0$$

$$2x = \cos x + 3$$

$$x = \frac{\cos x + 3}{2}$$

$$\text{i.e. } f(x) = \frac{1}{2} (\cos x + 3)$$

$$f'(x) = -\frac{\sin x}{2} < 1$$

So, iterative method can be applied to above equation.

→ Program 3

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
float f (float x)
```

```
{
```

```
float d;
```

```
d = (cos(x) + 3) / 2;
```

```
return (d);
```

```
}
```


∴ flag is name of variable.

```
void main ( )
```

```
{
```

```
float e, x0, x1, x2;
```

```
int i, n, flag = 1;
```

```
cout << "Simple iterative method" << endl;
```

```
cout << "Enter value of x0;"
```

```
cout << "Enter maximum no. of  
Iteration";
```

```
cin >> n;
```

```
cout << "Enter the value of pre-  
defined accuracy";
```

```
cin >> e;
```

```
x1 = f(x0);
```

```
cout << "Enter flag" << endl;
```

```
cin >> flag;
```

```
for ( i = 0 ; i < n ; i++ )
```

```
{
```

```
x2 = x1 - x0;
```

```
if (x2 < 0)
```

```
x2 = x2 * (-1)
```

```
if (x2 < e)
```

```
flag = 0;
```



```

else
{
    x0 = x1 ;
    x1 = f(x0) ;
}
cout << i << flag << "\t" << x1 << endl;
}
if (flag == 1)
    cout << "No convergence" << endl;
else
    cout << "root of eqn by"
    cout << "iterations, x1;"
    getch();
}

```

(III)
 Simpson's 1/3 rule:

for mula:

$$I = \frac{h}{3} [(f_0 + f_6) + 2(f_2 + f_4) + 4(f_1 + f_3 + f_5)]$$

Program:

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```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
float f ( float x)
```

```
{
```

```
float a;
```

```
a = expx;
```

```
return a;
```

```
}
```

```
void main()
```

```
{
```

```
float h, low, up, sum=0, n;
```

```
n = (up - low/h);
```

```
cout << n << endl;
```

```
sum = f(low) + f(up);
```

```
for (int i=1; i<n; i++)
```

```
{
```

```
if (i%2 == 0)
```

```
{
```

```
sum = sum + 2 * f ( low + i*h );
```

```
cout << "sum=" << sum << endl;
```

```
}
```



```
sum = sum * h/3;
```

```
cout << "Total result=" << sum << endl;
```

```
getch();
```

```
}
```

(I)

Damped Harmonic Oscillation:

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
void main()
```

```
{
```

```
float ta[1000], xa[1000], aa[1000],  
va[1000];
```

```
float t, h, tmax, x, v, a, k, m, b;
```

```
x = 0;
```

```
v = 50;
```

```
t = 0;
```

```
tmax = 5.5;
```

```
m = 1;
```

```
k = 1;
```


$b = 0.5;$

$g = 9.8;$

$\text{int ind} = 0;$

$\text{while } (t \leq t_{\max})$

{

$a = ((-k) * x) - (b * v) / m;$

$ta[\text{ind}] = t;$

$xa[\text{ind}] = x;$

$va[\text{ind}] = v;$

$aa[\text{ind}] = a;$

$x = x + v * h;$

$v = v + a * h;$

$t = t + h;$

$\text{ind} = \text{ind} + 1;$

}

$\text{cout} \ll \text{time} \ll t \ll \text{velocity} \ll t$

$\ll \text{acceleration} \ll \text{endl};$

$\text{cout} \ll ta[i] \ll t \ll xa[i] \ll t \ll$

$va[i] \ll t \ll aa[i] \ll \text{endl};$

$\text{getch}();$

}

Short Questions :-

(i)

Program :

```
float f (float p)
```

```
{
```

```
    P = m * v;
```

```
    K.E = 1/2 * m * v2;
```

```
    return p;
```

```
}
```

```
void main()
```

```
{
```

```
    float m, v, P, K.E;
```

```
    cout << "Enter the mass of particle";
```

```
    cin >> m;
```

```
    cout << "Enter the velocity of particle";
```

```
    cin >> v;
```

```
    P = m * v;
```

```
    K.E = 1/2 * m * v2;
```

```
    cout << "Kinetic Energy of particle <<
```

```
    "/n" << "momentum of particle << endl;
```

```
    getch();
```

```
}
```


(iv)

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
void main()
```

```
{
```

```
float theta, L, T, g = 9.8, Pi = 3.14;
```

```
cout << "Length" << "\t" << "Time" << endl;
```

```
for (L = 0.2; L <= 2; L = L + 0.05)
```

```
{
```

```
T = 2 * Pi * sqrt(L / g);
```

```
cout << L << "\t" << T << endl;
```

```
}
```

```
getch();
```

```
}
```

(v)

Area of circle = πr^2

Circumference of circle = $2\pi r$

Program:

```
#include <iostream.h>
```

```
#include <conio.h>
```



```
#include <math.h>
```

```
void main()
```

```
{
```

```
float x[7], A[7], C[7];
```

```
cout << "Enter x = ";
```

```
for (int i=0; i<7; i++)
```

```
{
```

```
cin >> x[i];
```

```
A[i] = 3.14 * x[i] * x[i];
```

```
C[i] = 2 * 3.14 * x[i];
```

```
}
```

```
cout << x[i] << "\t" << A[i] << "\t"
```

```
<< C[i] << endl;
```

```
getch();
```

```
}
```

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(vi)

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
    int a[4][3], i, j;
```

```
    cout << "Enter the elements of  
matrix" << endl;
```

```
    for (i=0; i<4; i++)
```

```
        for (j=0; j<3; j++)
```

```
        {
```

```
            cin >> a[i][j];
```

```
        }
```

```
    cout >> "You have entered a  
matrix as: << endl;
```

```
    for (i=0; i<4; i++)
```

```
    {
```

```
        for (j=0; j<3; j++)
```

```
        {
```

```
            cout << a[i][j] << endl;
```

```
        }
```

```
    }
```

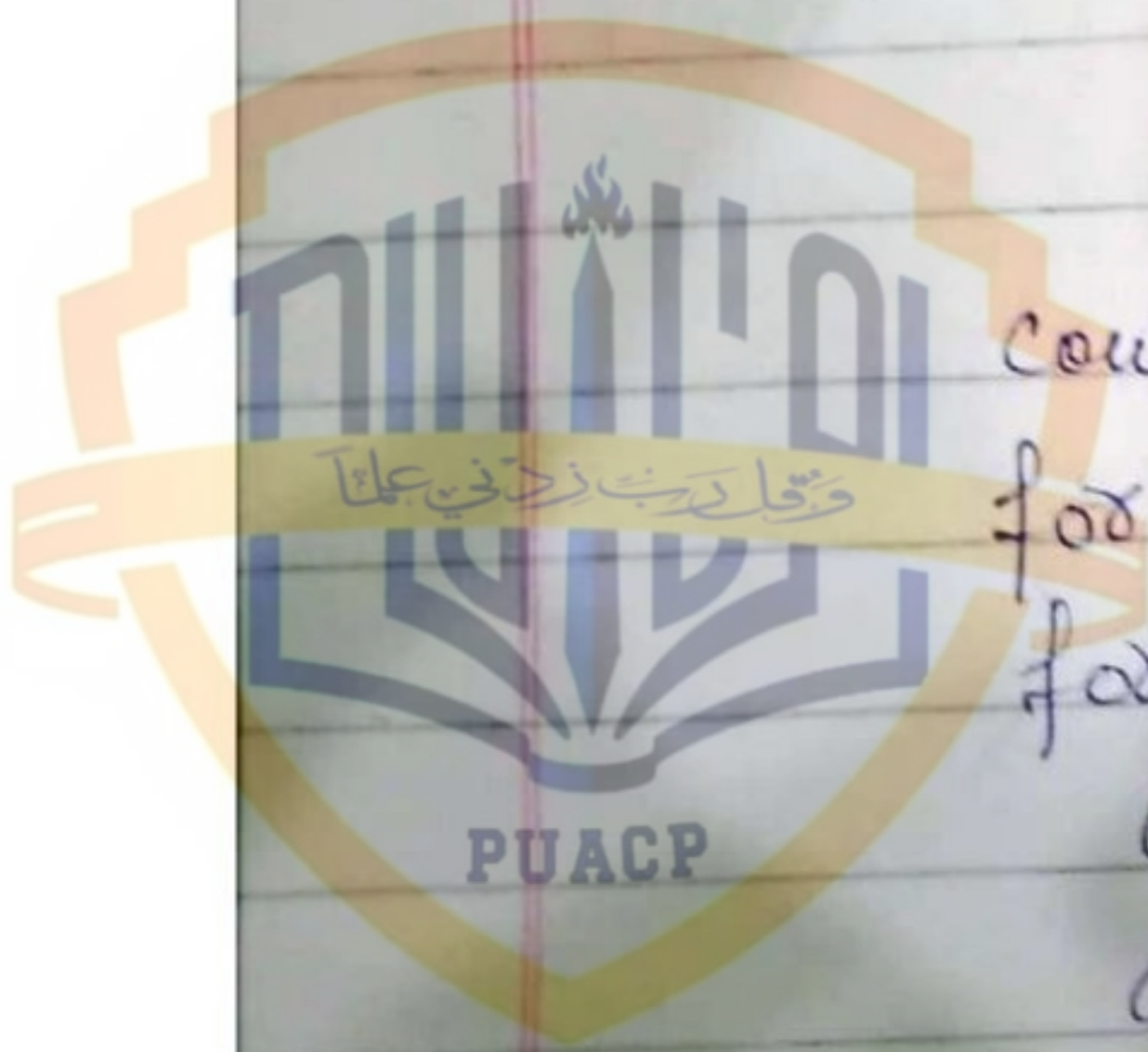
```
    getch();
```

```
}
```


(iii)

```
#include <iostream.h>
#include <conio.h>
#include <math.h>

void main()
{
    int A[3][3], B[3][3], C[3][3];
    cout << "Enter A=" << endl;
    for (int i=0; i<3; i++)
        for (int j=0; j<3; j++)
            cin >> A[i][j];
    cout << "A=";
    for (int i=0; i<3; i++)
    {
        for (int j=0; j<3; j++)
            cout << A[i][j] << endl;
    }
    cout << "Enter B=";
    for (int i=0; i<3; i++)
        for (int j=0; j<3; j++)
            cin >> B[i][j];
    cout << "B=";
```




```
for (int i=0; i<3; i++)
```

```
{  
    for (int j=0; j<3; j++)  
        cout << B[i][j];
```

```
    }  
for (int i=0; i<3; i++)  
for (int j=0; j<3; j++)  
{  
    C[i][j] = 0;
```

```
    for (int k=0; k<3; k++)  
        C[i][j] += A[i][k] * B[k][j]  
    cout << C[i][j] << endl;
```

// inverse of A and B

```
invC[i][j] = invA[i][k] * invB[k][j]
```

```
cout << "Inverse of matrix" <<  
"invA[i][k] * invB[k][j]" <<
```

```
invC[i][j] << endl;
```

```
getch();
```

```
}
```


(ii)

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
void main()
```

```
{  
    int a[3][3], b[3][3], i, j;
```

```
    cout << "Enter the matrix a:";
```

```
    for (i=0; i<3; i++)
```

```
    {  
        for (j=0; j<3; j++)
```

```
        {  
            cin >> a[i][j];
```

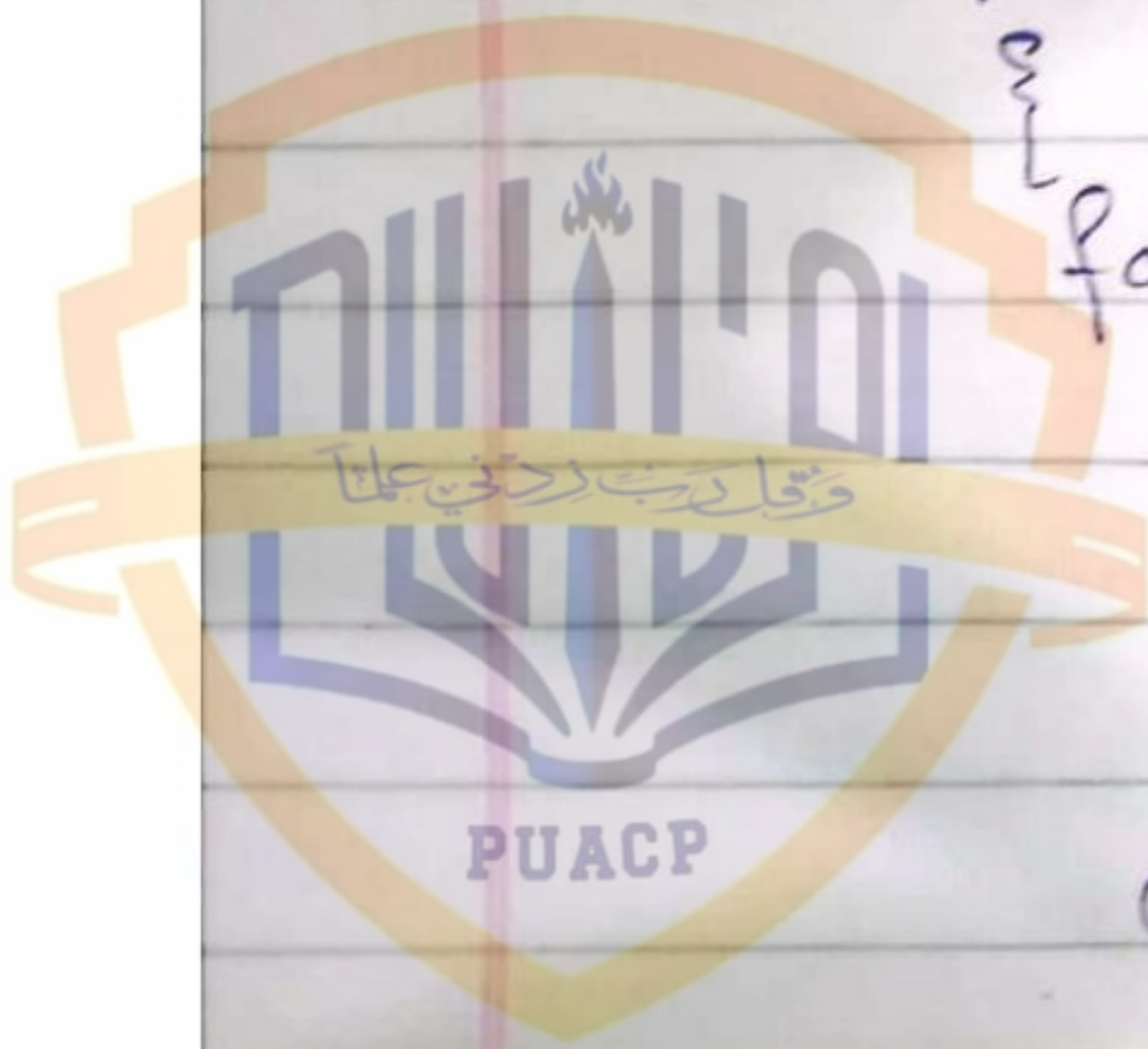
```
        }  
        cout << "matrix is=" << endl;
```

```
        for (i=0; i<3; i++)
```

```
        {  
            for (j=0; j<3; j++)
```

```
            {  
                cout << a[i][j] << " ";
```

```
            }  
            cout << endl;
```




```
cout << "In main diagonal elements  
are : " << endl;
```

```
for (i = 0; i < 3; i++)
```

```
{
```

```
    for (j = 0; j < 3; j++)
```

```
    {
```

```
        if (i == j)
```

```
            cout << a[i][j] << " ";
```

```
        }
```

```
    }  
    cout << endl;
```

```
}
```

```
cout << "Enter the elements  
of matrix b = ";
```

```
for (i = 0; i < 3; i++)
```

```
{
```

```
    for (j = 0; j < 3; j++)
```

```
    {
```

```
        cin >> b[i][j];
```

```
    }
```

```
cout << "matrix is = " << endl;
```

```
for (i = 0; i < 3; i++)
```

```
{
```



```

for (j=0; j<3; j++)
{
    cout << b[i][j] << " ";
}
cout << endl;

```

```

}
cout << "\n Main diagonal
elements are;" << endl;

```

```

for (i=0; i<3; i++)
{
    for (j=0; j<3; j++)
    {
        if (i == j)
            cout << b[i][j] << " ";
    }
    cout << endl;
}

```

```

}
c[i][j] = a[i][j] * b[i][j]

```

// diagonal entries of $a \times b$

```

cout << "diagonal elements of
a * b" << "\t" << c = a * b << endl;
getch();
}

```


2022 → short Q#2

by another method:

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
void main()
```

```
{
```

```
int A[3][3], B[3][3], C[3][3];
```

```
int i, j;
```

```
cout << "Enter elements of  
matrix A";
```

```
for (i=0; i<3; i++)
```

```
{
```

```
for (j=0; j<3; j++)
```

```
{
```

```
cin >> A[i][j];
```

```
}
```

```
}
```

```
cout << "Enter elements of  
matrix B";
```

```
for (i=0; i<3; i++)
```

```
{
```

```
for (j=0; j<3; j++)
```

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```

{
    cin >> B[i][j];
}
}

```

// Matrix Multiplication:

```

for (i=0; i<3; i++)

```

```

{
    for (j=0; j<3; j++)

```

```

    {
        C[i][j] = 0;

```

```

        for (int k=0; k<3; k++)

```

```

        {
            C[i][j] += A[i][k] * B[k][j];

```

```

        }

```

```

    }

```

```

}

```

cout << "Diagonal elements of matrix C:";

```

for (i=0; i<3; i++)

```

```

{

```

```

    cout << C[i][i] << " ";

```

```

}

```



getch (1);

